Himel Mallick, PhD, FASA **Last updated on: November 1, 2026**

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Summary

Data Scientist and Computational Biologist with two decades of experience in Statistics, Biostatistics, and AI/ML in academia and industry. Highly cited publication track record with over 50+ publications and widely used computational tools (10+ R packages). Specialized in Bayesian statistics, machine learning, and omics data science methodologies including multimodal integration, microbiome, single-cell, spatial transcriptomics, imaging, and digital pathology. Recipient of the IISA ECASDS Award, 2022 • 2023 Fellow of the ASA (FASA) • 2023 Elected Member of the ISI.

Education and Training

Postdoctoral Fellowship, Biostatistics, Harvard University and Broad Institute 2019
Ph.D., Biostatistics, University of Alabama at Birmingham (GPA: 4.0/4.0) 2015
M.Sc., Statistics, IIT Kanpur (Class Rank: 4) 2009
B.Sc., Statistics, University of Calcutta (First Class Honours) 2007

Experience

Tenure-track Faculty of Biostatistics, Population Health Sciences 04/2023 – Present
Adjunct Faculty of Statistics and Data Science 06/2023 – Present
Adjunct Faculty of Gastroenterology and Hepatology 11/2025 – Present
Cornell University

- As a tenure-track faculty member at Cornell, I am building a cross-disciplinary team of statisticians, computational biologists, and AI scientists, bringing together powerful techniques from AI/ML, statistics, systems biology, and omics data science to develop reverse translation solutions for target identification and biomarker discovery.
- I hold a joint appointment in the Division of Gastroenterology and Hepatology as well as the Department of Statistics and Data Science, where my primary goal is to mentor and collaborate with Cornell investigators.

Senior/Associate Principal Scientist, Biostatistics 03/2019 – 04/2023
Merck Research Laboratories

- Led multiple data science projects providing end-to-end bioinformatics and biostatistics support across all stages of biomarker discovery and development including assay quality control, drafting SAP, and performing statistical analyses
- Supported Merck's internal efforts in the single-cell and spatial transcriptomics initiatives for translational oncology studies
- Liaised with digital pathology scientists and bioinformaticians on crucial statistical analyses for both exploratory and clinical development purposes
- Employed machine learning classification to the discovery of clinically actionable biomarkers from integrated multi-omics profiles to enable better disease outcome prediction and patient stratification
- Secured funding to initiate and execute multiple academic collaborations to conduct research on various 5-year strategic initiatives specified by the organization
- Successfully hired and co-mentored 4 Ph.D. summer interns towards the development and implementation of sophisticated statistical and AI/ML approaches for enabling personalized medicine
- Published research across several disease areas (infectious diseases, oncology, and microbiology) that utilized innovative statistical methodology and systems biology techniques spanning a wide range of translational applications

Postdoctoral Researcher, Computational Biology and Bioinformatics 10/2015 – 03/2019
Huttenhower lab, Harvard University and Broad Institute

- Lead developer of MaAsLin 2, a R/Bioconductor package for associating microbial multi-omic data with arbitrarily complex clinical metadata ($> 50,000$ official downloads)
- Lead developer of MelonnPan, a tool to predict metabolite profiles from metagenomic sequencing data using concepts from machine learning and ecology, implemented in R (> 250 citations)
- Contributed to grant writing, manuscript preparation, interdisciplinary collaborations, and teaching and mentoring of graduate and undergraduate students and trainees

Intern, Biostatistics Summer 2015
Mayo Clinic
- Developed Bayesian adaptive trial designs for clinical trials utilizing surrogate endpoints in the presence of biomarkers

Research Assistant, Biostatistics 08/2010 – 04/2015
University of Alabama at Birmingham (UAB)
- Developed Bayesian machine learning methods for high-dimensional feature selection in personalized medicine applications
- Developed and validated risk prediction models for assessing short-term mortality in obese adults
- Conducted high-dimensional predictive modeling of zero-inflated count phenotypes to identify genetic susceptibility markers in Rheumatoid Arthritis patients

Intern, Biostatistics Summer 2014
Novartis Pharmaceuticals Corporation
- Developed novel Bayesian methods to conduct heterogeneity of treatment effect (HTE) analyses in phase III clinical trials

Intern, Biostatistics Summer 2013
University of Arkansas for Medical Sciences
- Developed methods for detecting maternal-fetal gene-gene interactions associated with obstructive heart defects in newborns from mother-offspring paired genetic data

Intern, Biostatistics Summer 2008
Indian Statistical Institute
- Performed research on non-linear statistical modeling of cross-sectional growth curve data by the Preece-Baines growth model

Computer Skills

Programming Languages:	C, C++, Python, Julia, SQL
Deep Learning:	Keras, PyTorch, TensorFlow
Statistical Software:	SAS, R, MATLAB, SPSS
Computing Platforms:	Windows, Macintosh, Linux
Applications:	MS Office, LaTeX, Adobe Illustrator, Shiny, Git

Notable Software

MaAsLin2 | **MelonnPan** | **Tweedieverse** | **BayesRecipe** | **IntegratedLearner**

Top 5 First-author Publications (Total = 60+, Citations = 13350, h-index = 26, i-10 index = 38)

Mallick et al. (2024). An Integrated Bayesian Framework for Multi-omics Prediction and Classification. *Statistics in Medicine* 43(5):983–1002. [Citations: 44]

Mallick et al. (2022). Differential Expression of Single-cell RNA-seq Data using Tweedie Models. *Statistics in Medicine* 41(18):3492-3510. [Citations: 48]

Mallick et al. (2021). Multivariable Association Discovery in Population-scale Meta-omics Studies. *PLoS Computational Biology* 17(11):e1009442. [Citations: 3,155]

Mallick et al. (2019). Predictive Metabolomic Profiling of Microbial Communities Using Amplicon or Metagenomic Sequences. *Nature Communications* 10(1):3136-3146. [Citations: 339]

Mallick and Yi (2014). A New Bayesian LASSO. *Statistics and Its Interface* 7(4):571–581. [Citations: 91]

Select Honors

2026 Ritu Banga Impact in Healthcare Research Award, Weill Cornell Medicine
2024 Rising Star Award, UAB National Alumni Society
2023 Elected Member, International Statistical Institute
2023 Fellow, American Statistical Association (ASA)
2022 Early Career Award in Statistics and Data Sciences (ECASDS), IISA
2018 IMS New Researcher Travel Award, IMS
2017 Best Paper Award, Science Unbound Foundation
2016 AMS-Simons Travel Grant, American Mathematical Society
2015 Best Student Paper Award, American Statistical Association
2014 Best Student Paper Award, International Indian Statistical Association
2014 Grants-in-aid of Research, Sigma Xi
2013 Naomi Fineberg Distinguished New Researcher Award, UAB
2012 Best Talk, UAB Graduate Students Research Day